Write a program to implement Logistic Regression (LR) algorithm in python

Code

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# -----------------------------
# Load Iris Dataset
# -----------------------------
iris = load_iris()

X = iris.data
y = iris.target

# Create DataFrame (Optional)
df = pd.DataFrame(X, columns=iris.feature_names)
df['Target'] = y

print("First 5 Records:\n")
print(df.head())

# -----------------------------
# Split Dataset
```

```
# ----------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# ----------------------------
# Feature Scaling
# ----------------------------
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# ----------------------------
# Train Logistic Regression Model
# ----------------------------
model = LogisticRegression(max_iter=200, random_state=42)
model.fit(X_train, y_train)

# ----------------------------
# Prediction
# ----------------------------
y_pred = model.predict(X_test)

# ----------------------------
# Results
# ----------------------------
print("\nActual Labels:")
print(y_test)
```

```python
print("\nPredicted Labels:")
print(y_pred)

print("\nAccuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))

# -----------------------------
# Predict New Sample
# -----------------------------
new_sample = [[5.1, 3.5, 1.4, 0.2]]
new_sample = scaler.transform(new_sample)

prediction = model.predict(new_sample)

print("\nNew Sample Prediction:")
print("Predicted Class:", iris.target_names[prediction[0]])
```

```
First 5 Records:

   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   Target
0       0
1       0
2       0
3       0
4       0

Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy: 1.0

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        19
  versicolor       1.00      1.00      1.00        13
   virginica       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45


New Sample Prediction:
Predicted Class: setosa
```